



Vantage

Portfolio Manager

Team 8 - Michael Taleb, Sasi Vattikuti, Katelyn Vo



Table of contents

01

Introduction

02

Application

03

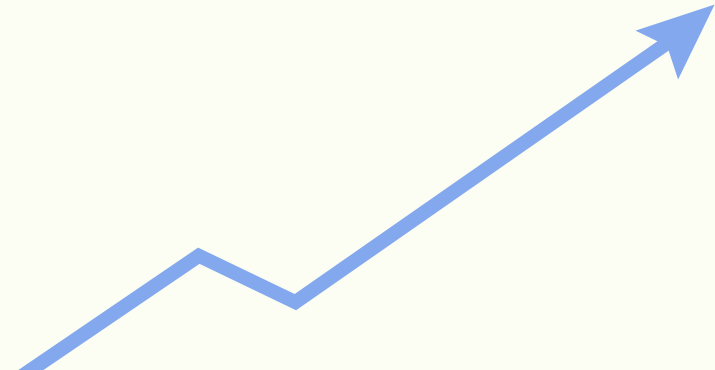
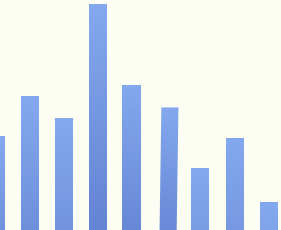
Challenges

04

Live Demo

05

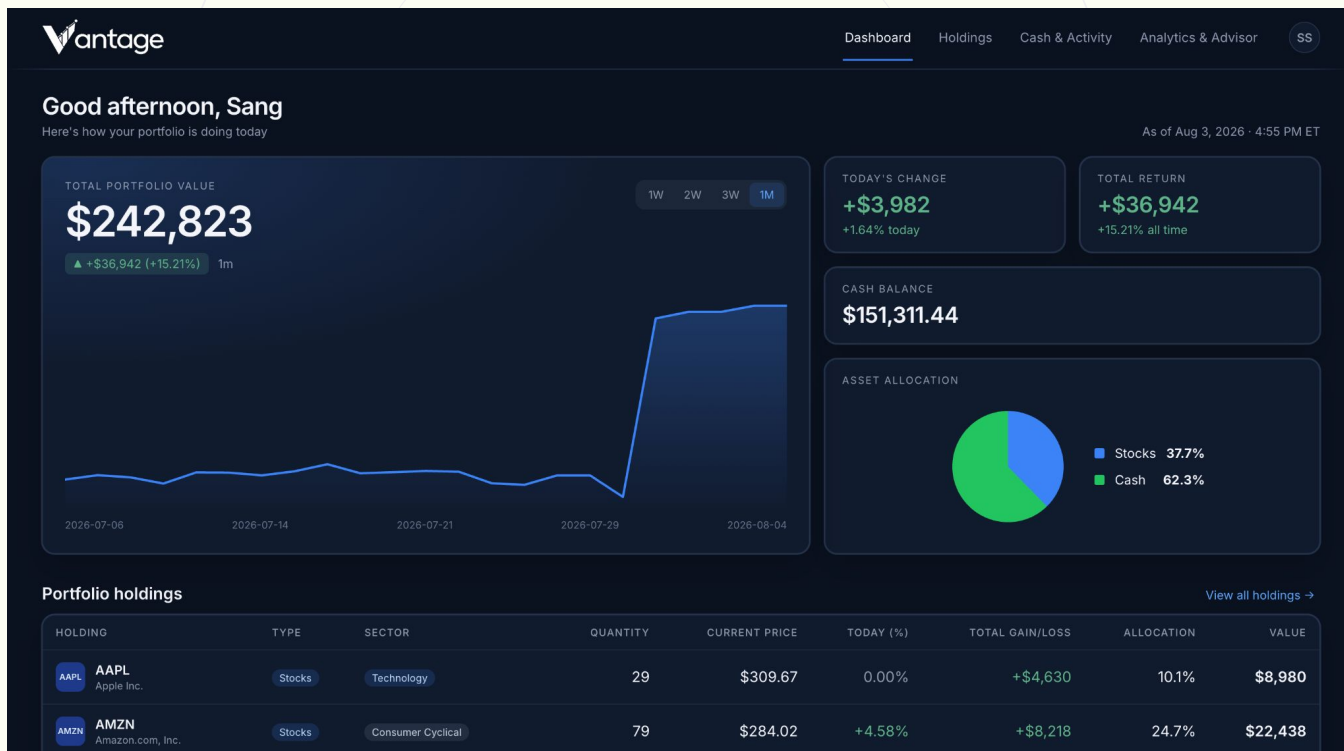
Q&A



The background features a light green gradient with various financial-themed elements. In the top left, there is a candlestick chart with white and blue bars. A blue line graph with an upward-pointing arrow starts from the left edge and trends upwards. In the top right, another candlestick chart is visible, with blue bars. Three 3D coins with dollar signs (\$) are scattered around: one large one in the top right, one medium one in the bottom left, and one small one in the middle right. At the bottom, there are two more line graphs; one on the left with blue bars and one on the right with a blue line graph and an upward arrow. In the center, there are three concentric circles. The text "01 Intro" is centered within these circles.

01 Intro

Meet Vantage



Features

Dashboard

- total value, performance chart, day gain / total return, asset & sector allocation

Holdings

- buy and sell stocks
- filter & sort
- drill into stock detail

Cash & Activity

- balance, deposit/withdraw, filterable transaction history

Quant Advisor

- AI-powered portfolio summary, chat, and forecast

Profile

- edit your display name/email

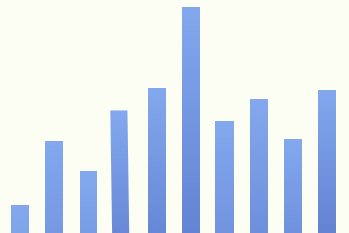
Live prices

- real-time updates via WebSocket, no refresh needed



Our Approach & Tech Stack

- Divided project into frontend (1 person) and backend (2 people) responsibilities
- **Frontend:** React, TypeScript, Vite, plain CSS
- **Backend:** Python, Flask, SQLAlchemy, MySQL, flask-sock (WebSockets)
- **External services:** yfinance (market data), Google Gemini API (AI advisor)
- **Workflow:** Git/GitHub



The background features a light cream color with various financial-themed elements. In the top left, there is a candlestick chart with white and blue bars. A blue line graph with an upward-pointing arrow starts from the left edge and trends upwards. In the top right, another candlestick chart is visible, with a large blue coin featuring a dollar sign (\$) floating below it. The bottom left shows a blue coin with a dollar sign. The bottom right has a blue line graph with an upward-pointing arrow. In the center, there are two concentric light gray circles. The text '02' is in a large blue font, and 'Application' is in a large dark blue font, both centered within the circles.

02 Application



Data Model

Tables

1.

User Assets

- Information about stocks user holds
- What price they were bought at

2.

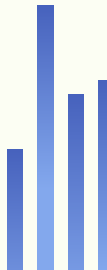
Portfolio History

- Auto Populated everyday
- Price of portfolio everyday for the past moth

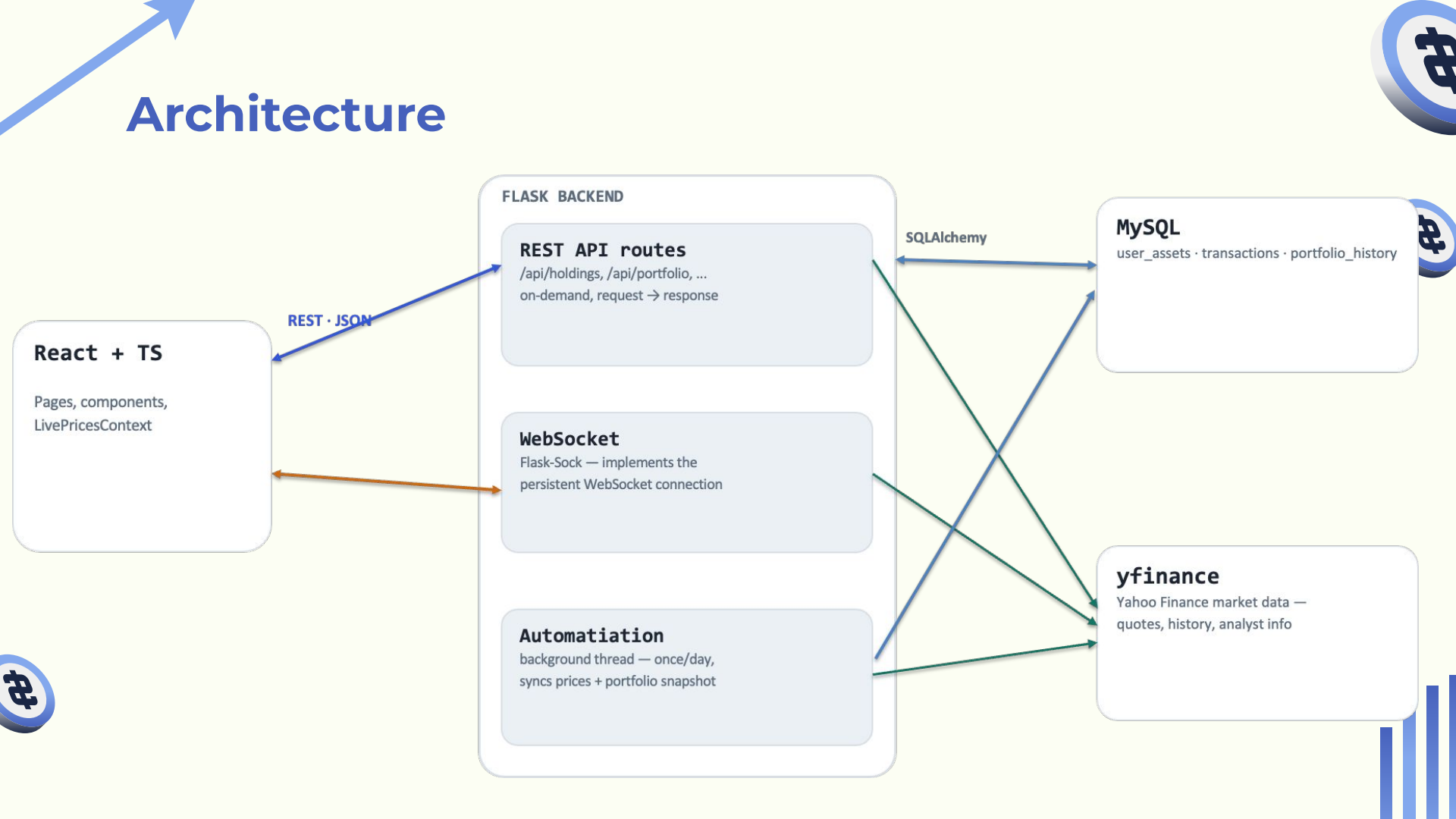
3.

Transactions

- History of buys and sells
- Prices and dates



Architecture





Analytics

Diversification Index

- *Uses the Herfindahl Index to measure how diversified a portfolio is across different sectors.*

Health Score

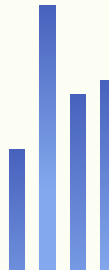
- *Takes volatility and the Diversification Index into account. To produced an overall score out of 100.*

Investment Efficiency

- *Measures how much return per unit risk (volatility) taken. Overall efficiency index.*

Monte Carlo Forecast

- *Uses portfolio drift and volatility to show possible future values*



The background features a light cream color with several faint, stylized financial charts. In the top left, there's a candlestick chart with white and blue bars. In the top right, a blue line chart trends upwards. In the bottom left, a blue bar chart shows fluctuating values. In the bottom right, another blue line chart trends upwards. Centered in the background is a target symbol consisting of three concentric circles with small dots on each circle's circumference.

03 Challenges



Challenges

Technical Challenges

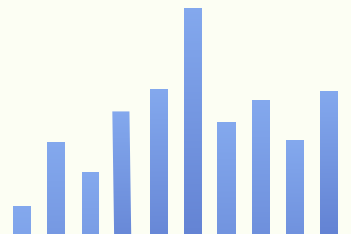
- Calculating the day change (opening price vs. yesterday's close)
- Calculating the gain/loss current value (using current value & daily change vs starting value)

Design Challenges

- Created Stock Table

If we had more time...

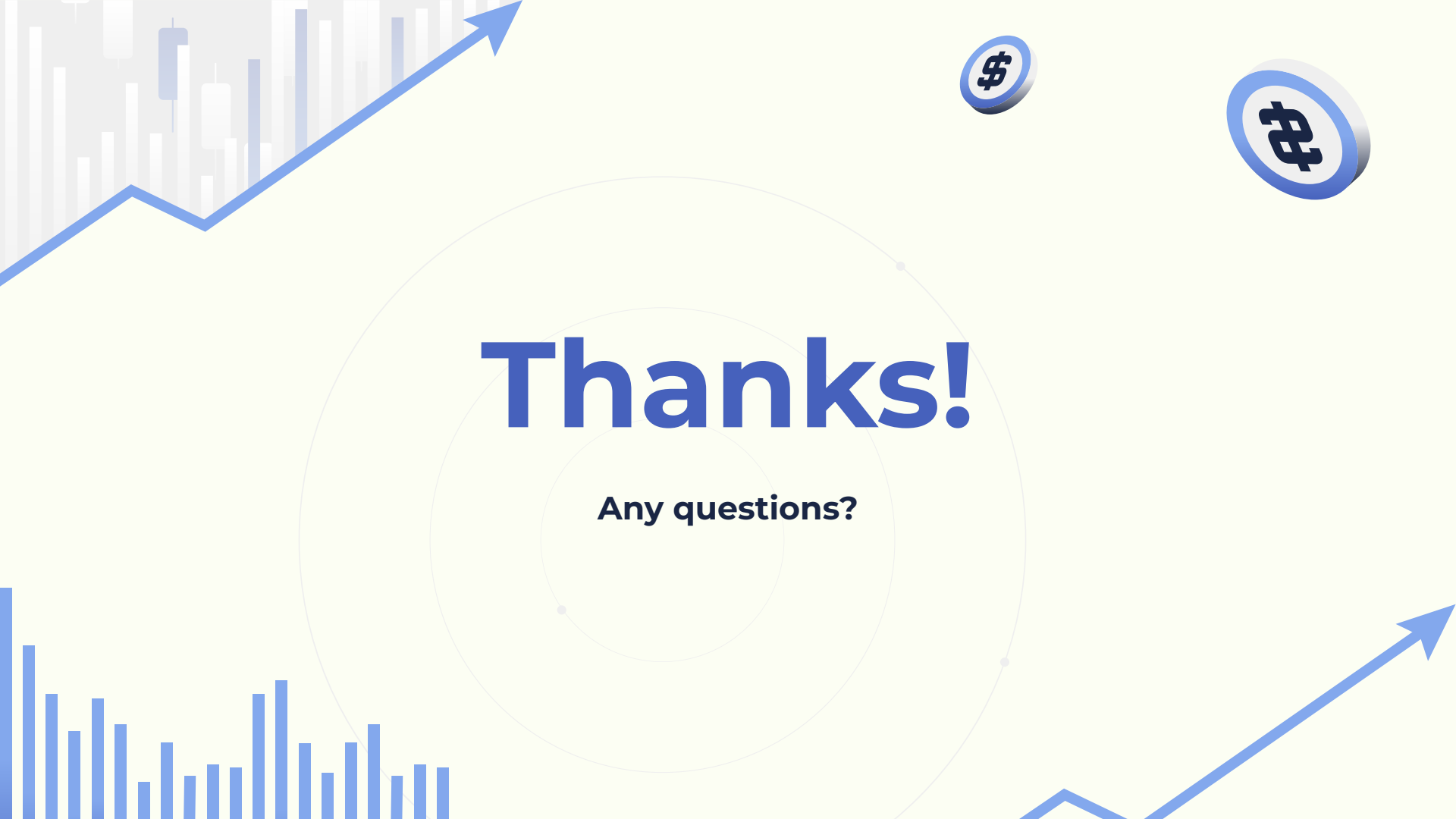
- Enhance investment features by adding other asset classes (bonds, ETFs, etc.)
- Optimize performance



The background features a light cream color with faint, stylized financial charts. On the left, a blue line graph trends upwards. On the right, a blue bar chart shows fluctuating values. At the bottom, another blue line graph trends upwards. In the center, a target graphic consists of three concentric circles with small dots on each circle's circumference.

04 Live Demo



The background features a light cream color with several financial-themed graphics. In the top-left corner, there is a grey bar chart and a blue line graph with an upward-pointing arrow. In the top-right corner, there are two blue circular coins with white dollar signs, one slightly larger than the other. In the bottom-left corner, there is a blue bar chart. In the bottom-right corner, there is a blue line graph with an upward-pointing arrow. In the center, there are three concentric circles in a light grey color.

Thanks!

Any questions?